

Pixel Relationships and Discrete Spatial Foundations in Digital Vision Systems

Digital image processing treats an image as a discrete two-dimensional function $f(x, y)$ sampled across an integer spatial lattice $\mathbb{Z}^2$, where spatial coordinates define discrete sensor elements and function amplitudes represent quantized radiance values¹. Unlike continuous Euclidean space $\mathbb{R}^2$, where spatial proximity and boundary continuity are governed by continuous differential topology, discrete digital grids require explicit combinatorial definitions of spatial association³. The fundamental operations of computer vision—edge tracing, region segmentation, morphological filtering, and distance transforms—rest upon how individual pixels interact with their spatial neighbors¹. Establishing these interactions requires defining neighborhood geometry, formulating mathematical criteria for adjacency, resolving digital topological paradoxes, tracing connected components, and computing distance transforms across discrete coordinate networks³.

The relationships between pixels operate at the intersection of discrete geometry and computational graph theory⁹. Without mathematically sound connectivity and metric rules, vision pipelines fail: morphological filters leak through object corners, connected-component labeling algorithms partition continuous objects into spurious disjoint fragments, contour-following routines enter non-terminating loops, and spatial transforms compute distorted distance representations³. Resolving these issues requires modeling both spatial adjacency (geometric proximity on the grid) and intensity similarity (shared luminance or spectral values)⁷.

The following technical presentation curriculum provides an exhaustive specification of discrete pixel relationships. Following the pedagogical architecture of graduate computer vision seminars—progressing from formal mathematical definitions to conceptual translations, coordinate grid layouts, structural comparison tables, algorithmic execution flows, and real-world system pipelines—this monograph synthesizes theoretical principles with production-grade vision mechanics¹.

Spatial Neighborhood Geometry across Discrete Coordinate Lattices

Formal Definition

Let an image be represented by a two-dimensional discrete array $f(x, y)$, where $(x, y) \in \mathbb{Z}^2$ denote integer spatial coordinates¹. For an arbitrary pixel p located at coordinates (x, y) , its spatial neighborhood is defined by subsets of adjacent coordinate points determined by orthogonal or diagonal displacements⁷.

The 4-neighborhood of p , denoted as $N_4(p)$, comprises the set of four orthogonal horizontal and vertical neighbors located at unit coordinate distance from (x, y) :

$$N_4(p) = \{(x + 1, y), (x - 1, y), (x, y + 1), (x, y - 1)\}$$

[cite: 5, 7, 8]

The diagonal neighborhood of p , denoted as $N_D(p)$, comprises the set of four corner neighbors located at a Euclidean displacement of $\sqrt{2}$ from (x, y) :

$$N_D(p) = \{(x + 1, y + 1), (x + 1, y - 1), (x - 1, y + 1), (x - 1, y - 1)\}$$

[cite: 5, 7, 8]

The 8-neighborhood of p , denoted as $N_8(p)$, is the union of the orthogonal and diagonal neighborhood sets:

$$N_8(p) = N_4(p) \cup N_D(p)$$

[cite: 2, 5, 8]

Spatial Coordinate Layout of Pixel Neighborhoods		
$(x - 1, y - 1)$	$(x, y - 1)$	$(x + 1, y - 1)$
$(x - 1, y)$	$p(x, y)$	$(x + 1, y)$
$(x - 1, y + 1)$	$(x, y + 1)$	$(x + 1, y + 1)$

Conceptual Translation

Every pixel inside a digital image sits on a uniform square grid¹. The pixel communicates with its surrounding environment through two primary groups: its four direct orthogonal compass neighbors (North, South, East, and West), which make up its 4-neighborhood, and its four diagonal corner neighbors (North-East, North-West, South-East, and South-West), which make up its diagonal neighborhood². When all eight surrounding squares are considered together,

they form the 8-neighborhood⁵.

Boundary Truncation Mechanics and Image Border Conditions

For an image array of size $M \times$ with row indices $x \in [0, M - 1]$ and column indices $y \in [0, N - 1]$, pixels located along the perimeter of the grid cannot support complete spatial neighborhoods². If a coordinate $(x \pm 1, y \pm 1)$ falls outside the domain $[0, M - 1] \times [0, N - 1]$, that point is undefined within the coordinate space².

Pixel Spatial Location	Valid N4(p) Count	Valid ND(p) Count	Valid N8(p) Count	Boundary Truncation Example
Interior Grid Element	4	4	8	Coordinates remain fully within array boundaries
Boundary Edge (Non-Corner)	3	2	5	At $x =$, coordinate $(x -$ is undefined ²
Corner Element	2	1	3	At $(0, 0)$, only $(0, 1)$, $(1, 0)$, and $(1, 1)$ exist ²

In practical computer vision implementations, boundary truncation is managed using four standard boundary extension strategies: constant zero-padding (clamping outside coordinates to zero), border replication (repeating the nearest boundary value), periodic extension (toroidal wrapping around edges), or symmetric reflection across the boundary margin¹³.

Adjacency Criteria and the Mechanics of Mixed Connectivity

Mathematical Formulation of Adjacency and Similarity Sets

Spatial adjacency alone is insufficient to establish connectivity; pixels must also satisfy a

similarity constraint⁷. Let V denote the set of intensity values used to define similarity⁷. For binary imagery, $V = \{1\}$ isolates foreground objects from background regions⁷. In 8-bit grayscale imagery, V can be defined as an arbitrary continuous or discrete intensity subset, such as $V = \{128, 129, \dots, 255\}$, representing a specific luminance range⁵.

Three discrete adjacency classes govern digital lattices:

- 4-Adjacency: Two pixels p and q with values from V are 4-adjacent if $q \in N_4(p)$ ⁵.
- 8-Adjacency: Two pixels p and q with values from V are 8-adjacent if $q \in N_8(p)$ ⁵.
- Mixed Adjacency (m -Adjacency): Two pixels p and q with values from V are m -adjacent if:
 1. $q \in N_4(p)$, or
 2. $q \in N_D(p)$ and the intersection set $N_4(p) \cap N_4(q)$ contains no pixels whose values are from V ⁵.

Structural Taxonomy of Pixel Adjacency Models
Adjacency Type
4-Adjacency
8-Adjacency
Mixed (m -) Adjacency

Conceptual Translation

Being adjacent on the grid does not make two pixels connected; they must also share the same visual class or grayscale intensity⁷. If two active pixels touch along an edge, they are 4-connected⁷. If they touch only at a corner, they can be 8-connected⁷.

Standard 8-connectivity introduces topological ambiguity because it allows two pixels to connect simultaneously through two routes: directly through their shared corner, and indirectly

through their shared orthogonal neighbors¹⁷. Mixed connectivity (m -connectivity) resolves this conflict: it allows a diagonal connection if and only if the orthogonal pathways between those pixels are inactive³.

Resolving Topological Ambiguities through Mixed Adjacency

The primary objective of m -connectivity is the elimination of multiple concurrent paths connecting the same pair of pixels¹⁷. When evaluating an 8-connected lattice, redundant pathways form digital loops that introduce ambiguities into boundary tracing and distance calculations³.

Consider a 3×3 binary image patch where the similarity set is defined as $V = \{1\}$:

Matrix Coordinate	Column 0	Column 1	Column 2
Row 0	0	1 _(q)	1 _(s)
Row 1	0	1 _(p)	0
Row 2	0	0	1

Let pixel p reside at coordinate $(1, 1)$ with value 1, pixel q reside at $(0, 1)$ with value 1, and diagonal neighbor s reside at $(0, 2)$ with value 1⁷:

- Pixels $p(1, 1)$ and $q(0, 1)$ are 4-adjacent because $q \in N_4(p)$ and both intensity values belong to V ⁷.
- Evaluating the relationship between $p(1, 1)$ and its diagonal neighbor $s(0, 2)$ reveals that their shared orthogonal neighborhood is $N_4(p(1, 1)) \cap N_4(s(0, 2)) = \{(0, 1), (1, 2)\}$ ⁵. Coordinate $(0, 1)$ contains pixel q with value $1 \in V$.
- Under 8-connectivity, pixel $p(1, 1)$ connects directly to $s(0, 2)$ along the diagonal step, while simultaneously connecting to $s(0, 2)$ via the two-step orthogonal path $p(1, 1) \rightarrow q(0, 1) \rightarrow s(0, 2)$ ⁷. This introduces two concurrent paths of lengths 1 and 2, forming an ambiguous closed triangle¹⁷.
- Under m -connectivity, the rule dictates that because $N_4(p) \cap N_4(s)$ contains a pixel with value $1 \in V$, the diagonal connection between $p(1, 1)$ and $s(0, 2)$ is

eliminated⁵. The only valid path remaining is the unambiguous orthogonal sequence $p(1, 1) \rightarrow q(0, 1) \rightarrow s(0, 2)$. This eliminates redundant loops and enforces well-defined digital boundaries⁸.

Discrete Jordan Curve Paradox and Dual Connectivity Frameworks

Topological Boundary Failure in Uniform Adjacency Models

In classical continuous topology $\mathbb{R}^2$, the Jordan Curve Theorem states that every simple, closed, non-self-intersecting curve divides the plane into exactly two disconnected components: a bounded interior and an unbounded exterior, with the curve acting as their shared boundary³. When this continuous space is sampled onto a discrete integer grid $\mathbb{Z}^2$, this topological property fails if identical connectivity rules are applied uniformly across both foreground and background sets³.

Continuous R2 vs. Discrete Z2 Topological Invariants
Topological Dimension
Space Nature
Boundary Separation
Boundary Crossing
Complement Adjacency

The Connectivity Paradox

Azriel Rosenfeld identified the digital connectivity paradox: if identical adjacency is enforced across both foreground objects and background regions, topological contradictions occur³.

Uniform 8-Connectivity Paradox (Diagonal Leakage)	Uniform 4-Connectivity Paradox (Background Disconnection)
An 8-connected digital curve loops diagonally across	A 4-connected digital curve encloses an interior pixel at $(1, 1)$ using orthogonal

$(0, 1), (1, 0), (2, 1), (1, 2)$. Foreground forms a closed boundary under 8-connectivity ⁷ .	neighbors: $(0, 1), (1, 0), (2, 1), (1, 2)$ ⁷ .
Background pixel $(0, 0)$ attempts to connect to interior center pixel $(1, 1)$ ³ .	Background pixel $(0, 0)$ attempts to connect to exterior background pixel $(2, 2)$ ³ .
If the background is also 8-connected, it steps diagonally from $(0, 0)$ directly into $(1, 1)$ between the boundary pixels without intersecting any foreground pixel ³ .	Under 4-connectivity, the exterior background cannot step diagonally across the corners of the object curve ³ .
Topological failure: The closed curve fails to separate its interior from the exterior (diagonal leakage) ³ .	Topological failure: A single closed object divides the surrounding background into multiple disconnected components ³ .

Demonstration of Diagonal Background Leakage		
B_{ext}	F_{obj}	B_{ext}
F_{obj}	B_{int}	F_{obj}
B_{ext}	F_{obj}	B_{ext}

In the configuration above, F_{obj} denotes foreground boundary pixels, B_{ext} represents exterior background pixels, and B_{int} represents an interior background pixel³. Under uniform 8-connectivity, an exterior pixel at B_{ext} connects diagonally to B_{int} through the corner gap between adjacent F_{obj} pixels, causing the interior and exterior regions to merge³.

The Dual Connectivity Solution and Boundary Invariants

To preserve the discrete analogue of the Jordan Curve Theorem, digital topology mandates dual connectivity³. If the foreground object S is evaluated under k -connectivity, its

background complement $\bar{S}$ must be evaluated under $\bar{k}$ -connectivity, where $(k, \bar{k}) \in \{(4, 8), (8, 4)\}$ ³. Under this framework, 4-connected boundaries enclose 8-connected backgrounds, and 8-connected boundaries enclose 4-connected backgrounds³. Alternatively, applying m -connectivity across both sets resolves the conflict by removing the intersecting diagonal paths³.

Milestones in Digital Topology and Grid Geometry
Period / Year
1966–1970
1984–1986
1985
1996

Digital Paths, Boundary Topologies, and Component Extraction Algorithms

Path Formulations and Region Topology

A digital path from pixel p at (x, y) to pixel q at (s, t) is a sequence of distinct pixels:

$$(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$$

where $(x_0, y_0) = (x, y)$, $(x_n, y_n) = (s, t)$, and each consecutive pair of pixels (x_i, y_i) and (x_{i-1}, y_{i-1}) are adjacent for all $1 \leq i \leq n$ ⁵. The integer n defines the path length⁷. If $(x_0, y_0) = (x_n, y_n)$ with $n \geq 1$, the path forms a closed path⁷. Paths are classified as 4-paths, 8-paths, or m -paths based on the adjacency metric enforced across the sequence⁷.

Let S represent a subset of pixels in an image. Two pixels $p, q \in S$ are connected in S if

there exists a path between them consisting entirely of pixels belonging to S ⁷. For any pixel $p \in S$, the set of all pixels in S connected to p constitutes a connected component of S ². If S contains only one connected component, S forms a connected region⁷.

Boundary, Edge, and Region Topology
Topological Structure
Inner Boundary (B_{in})
Outer Boundary (B_{out})
Hole (Interior Void)

Two-Pass Equivalence Resolution and Suzuki-Abe Border Following
 Connected-component labeling (CCL) transforms a binary pixel array into a symbolic label map, assigning unique integer identifiers to disjoint connected foreground regions¹⁰. The two-pass algorithm (Hoshen-Kopelman / Rosenfeld-Pfaltz) achieves near-linear computational efficiency by separating spatial scanning from equivalence resolution¹⁰.

Two-Pass Connected-Component Labeling Execution Stages
Processing Stage
Pass 1: Scan & Label
Intermediate: Resolve
Pass 2: Remap

Causal Neighbor Masks for First-Pass Scanning
Connectivity Model

4-Connectivity
8-Connectivity

During Pass 1, if an active foreground pixel $p(x, y)$ touches labeled causal neighbors with distinct identifiers (for example, a North neighbor with label 2 and a West neighbor with label 5), a label conflict occurs¹⁰. The algorithm assigns $p(x, y) = \min(2, 5) = 2$ and merges labels 2 and 5 in the disjoint-set data structure¹⁰. Rather than processing dense pixel interiors, the Suzuki-Abe border following algorithm traces outer boundaries and hole boundaries directly¹⁴. Scanning the image line by line until an unvisited transition occurs ($0 \rightarrow 1$ for an outer border, or $1 \rightarrow 0$ for a hole border), the algorithm initiates a border-following walk around the boundary contour until it returns to the starting pixel¹⁴. The algorithm assigns hierarchical relationships, establishing which holes are nested within which outer objects, producing an organized topological scene tree without requiring morphological filling operations¹⁴.

Comparative Taxonomy of Region Extraction Paradigms
Algorithmic Paradigm
Classical Recursive Flood Fill
Two-Pass with Union-Find (SUF)
Suzuki-Abe Border Following

Distance Metrics and Chamfer Transform Approximations

Metric Space Axioms and Classical Distance Formulations

A distance function D between two pixels $p(x, y)$ and $q(s, t)$ is a real- or integer-valued mapping that forms a metric space if and only if it satisfies four fundamental axioms for all pixels $p, q, z \in \mathbb{Z}^2$:

1. Non-negativity: $D(p, q) \geq 0$ ⁸.
2. Identity of Indiscernibles: $D(p, q) = 0 \iff p = q$ ⁸.
3. Symmetry: $D(p, q) = D(q, p)$ ⁸.
4. Triangle Inequality: $D(p, z) \leq D(p, q) + D(q, z)$ ⁶.

Digital image processing standardizes on three coordinate distance metrics⁵:

The Euclidean Distance (D_e):

$$D_e(p, q) = \sqrt{(x - s)^2 + (y - t)^2}$$

[cite: 5, 8] D_e measures true straight-line geometric distance⁵. The locus of points satisfying $D_e(p, q) \leq r$ forms a circular disk centered at (x, y) ⁸. While rotationally invariant, D_e outputs non-integer floating-point values requiring square root operations¹¹.

The City-Block / Manhattan Distance (D_4):

$$D_4(p, q) = |x - s| + |y - t|$$

[cite: 5, 8] D_4 measures distance assuming movement is restricted strictly to orthogonal 4-neighbor steps⁵. The locus of points satisfying $D_4(p, q) \leq r$ forms a diamond (a square rotated at 45°) centered at (x, y) ⁸.

The Chessboard / Chebyshev Distance (D_8):

$$D_8(p, q) = \max(|x - s|, |y - t|)$$

[cite: 5, 8] D_8 allows unrestricted movement across all 8 neighbors, assigning orthogonal and diagonal steps the same unit cost⁵. The locus of points satisfying $D_8(p, q) \leq r$ forms an axis-aligned square centered at (x, y) ⁸.

Geometric Contours of Digital Distance Metrics (r=1)

City-Block Metric (D_4):	Chessboard Metric (D_8):
$\begin{bmatrix} \cdot & 1 & \cdot \\ 1 & p & 1 \\ \cdot & 1 & \cdot \end{bmatrix}$	$\begin{bmatrix} 1 & 1 & 1 \\ 1 & p & 1 \\ 1 & 1 & 1 \end{bmatrix}$

Geometric Characteristics of Metric Contours
Metric Formulation
Euclidean (D_e)
City-Block (D_4)
Chessboard (D_8)

Integer Chamfer Approximations and Metric Violations in Mixed Distance

Computing exact Euclidean distances across large discrete images requires floating-point operations¹¹. To improve performance while minimizing geometric distortion, Gunilla Borgefors introduced Chamfer distance transforms, which approximate Euclidean metrics using integer convolution kernels⁴.

Chamfer Distance Kernel Taxonomy
Chamfer Kernel Class
City-Block Metric (D_4)
Chessboard Metric (D_8)

Borgefors 3-4 Chamfer
Borgefors 5-7-11 Chamfer

A distance transform takes a binary image of foreground objects and background points and assigns to each foreground pixel the metric distance to its nearest background pixel¹⁵. Using the Borgefors 3-4 mask, this transform is computed using a two-pass sequential raster scan:

Chamfer Distance Transform Two-Pass Propagation
Scan Pass
Forward Pass
Backward Pass

The mixed distance (D_m) is defined as the shortest path length between two pixels under m -connectivity⁷. Unlike D_e , D_4 , and D_8 , D_m is path-dependent and relies on the intensity values of neighboring pixels rather than coordinate positions alone⁸.

Because routing depends on intermediate pixel values, D_m can violate the Triangle Inequality:

$$D_m(p, z) \leq D_m(p, q) + D_m(q, z) \quad (\text{Fails in specific configurations})$$

[cite: 8, 30]

Coordinate Point	Grid Coordinate (x,y)	Pixel Value	Spatial Context
Point p	(0, 0)	1	Origin point
Point q	(1, 1)	1	Diagonal neighbor to p

Point z	$(1, 2)$	1	Orthogonal neighbor to q
Context Pixel r	$(0, 1)$	1	Shared orthogonal neighbor in $N_4(p) \cap N_4(q)$

In this configuration, evaluating the path between $p(0, 0)$ and $q(1, 1)$ reveals that their shared orthogonal neighbor $r(0, 1)$ contains a 1⁸. Under m -connectivity, the direct diagonal connection between p and q is eliminated⁵. The path must route through r : $p(0, 0) \rightarrow r(0, 1) \rightarrow q(1, 1)$, giving a path length of $D_m(p, q) = 2$ ⁸. The distance from $q(1, 1)$ to $z(1, 2)$ is direct: $D_m(q, z) = 1$ ⁸. The sum of intermediate distances is $D_m(p, q) + D_m(q, z) = 2 + 1 = 3$ ⁸.

However, evaluating the direct path from $p(0, 0)$ to $z(1, 2)$ reveals that an alternate route may circumvent the intermediate blocking condition entirely, altering the path length⁸. If neighboring background values fluctuate, $D_m(p, q)$ changes even when the coordinate positions of p and q remain fixed⁸. Because distance depends on surrounding pixel values rather than spatial coordinates alone, D_m does not form a true mathematical metric⁸.

Vision System Architectures and Industrial Implementation Pipelines

The practical value of pixel relationship theory is demonstrated when integrating these operations into end-to-end computer vision systems¹. From early space exploration imaging by NASA's Ranger 7 to contemporary industrial systems, spatial topology serves as the operational foundation for visual scene understanding¹.

Automated Histopathological Cell Segmentation Pipeline

In computational pathology, clinical diagnostics depend on segmenting, counting, and profiling red and white blood cells from brightfield microscopy scans¹. Touching and overlapping cells must be separated without introducing artificial fragmentation²⁶.

Operational Stage	Processing Step	Pixel Relationship Mechanics	Concrete Computational State
1. Acquisition	Multi-sensor micro-array capture	Discrete sampling onto a CMOS sensor grid; Bayer pattern CFA demosaicing ¹ .	Continuous specimen digitized into an $M \times$ discrete matrix $f(x, y)_1$.
2. Enhancement	Adaptive local contrast adjustment	Convolution across local N_8 windows to enhance signal-to-noise ratio ¹ .	Contrast-enhanced image with sharpened cell membrane boundaries ¹ .
3. Binarization	Otsu global intensity thresholding	Intensity similarity set definition: $V = \{g \in [0, 255]$ ² .	Binary foreground where cellular regions equal 1, background equals 0 ² .
4. Distance Transform	Borgefors 3-4 Chamfer Transform	Two-pass raster propagation computing metric distance to background ¹¹ .	Grayscale distance map where cell centers form local intensity maxima ³¹ .
5. Segmentation	Marker-controlled watershed	Gradient descent over inverted distance map using $(8, 4)$ dual connectivity ³ .	Separation of touching cells along contact boundaries ²⁶ .
6. Labeling	Two-pass CCL with Union-Find	Raster scan evaluating causal masks; resolves label equivalences via disjoint sets ¹⁰ .	Labeled symbolic image where every cell possesses a unique integer ID ¹⁰ .

7. Description	Suzuki-Abe border following	Traces 8-connected external cell borders and 4-connected nuclear holes ¹⁴ .	Vectorized boundary contours, circularity, perimeter, and area metrics ¹⁴ .
8. Classification	Morphological diagnosis	Morphological metrics compared against a pathological knowledge base ¹ .	Cell classification (e.g., distinguishing neutrophils from lymphocytes) ¹ .

Semiconductor Defect Inspection and Medial Axis Skeletonization

In automated semiconductor manufacturing, printed silicon wafers must be continuously verified for circuit continuity, short-circuit bridges, and trace thinning¹.

Inspection Phase	Technical Implementation	Core Neighborhood Mechanics	Operational Function
Surface Mapping	Borgefors 5-7-11 Chamfer Transform	Large $5 \times$ window propagation yielding 2.02% maximum Euclidean error ¹¹ .	Encodes distance of circuit interiors from conductive trace boundaries ³¹ .
Medial Axis Extraction	Centers of maximal disks extraction	Point p retained if no Chamfer disk centered elsewhere covers its disk ³² .	Identifies central structural ridges of conductive traces ³² .
Morphological Thinning	Topology-preserving thinning	Prunes non-essential pixels using m -connectivity rules ³³ .	Reduces traces to single-pixel-wide skeletons without introducing loops ³ .
Graph Topology	Neighborhood	Evaluates active	Detects circuit

Audit	order analysis	neighbor counts: $ N_8(p) =$ (endpoints), $ N_8(p) \geq$ (junctions) ²⁶ .	open-circuit breaks and unwanted short-circuit bridges ²⁶ .
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Analyzing digital images on discrete spatial grids requires adhering to the topological and metric constraints of local pixel neighborhoods³. Because continuous geometry does not map directly onto an integer lattice, properties such as boundary closure, path uniqueness, and metric isotropy must be explicitly enforced³. Reliable vision systems depend on matching operators to their underlying topological constraints: pairing $(4, 8)$ dual connectivity to preserve the Jordan Curve Theorem, using m -connectivity to eliminate redundant paths, applying Union-Find disjoint sets for efficient component labeling, and employing Chamfer integer kernels to balance computational speed with metric accuracy³. These pixel relationships form the foundational link between raw sensor arrays and structured computer vision systems¹.

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